RESEARCH ON ATTENTION-BASED MULTIGRANULAR REMOTE SENSING IMAGE EXTREME OBJECT DETECTION ALGORITHM

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Abstract:

The detection of small objects and high aspect ratio objects is a challenge for remote sensing image object detection. In this paper, we refer to these objects as extreme objects. The Attention-Based Multigranular Extreme Object Detection Network (AMEODNet) is proposed to address these problems. It consists of two parts. The first part is a multi-granularity information fusion module. The second part is an extreme object extraction section. The proposed method is tested on the NWPU VHR-10 datasets. The results are compared with mainstream object detection algorithms to verify the effectiveness of the proposed method.

Keywords:

Extreme Object Detection; Attention Mechanism; Multigranular; Remote Sensing Image

1. Introduction

Remote sensing image object detection is a fundamental task in processing high-resolution optical remote sensing images. It aims to identify the category and location of objects. This method is commonly used in military and civil applications. However, object detection is challenging due to multiple scales and different shapes of objects, complex background interference, high object density, and noise effects. Small objects consist of relatively few pixels and have few discerning features, resulting in missed detections. It is difficult to capture the global information of objects with high aspect ratio during feature extraction. The detection difficulty is exacerbated by relatively small objects with a high aspect ratio.

Commonly used object detectors are improvements of manual methods and utilise convolutional neural networks (CNNs), resulting in significant progress in object detection in recent years. Classical object detection algorithms based on deep learning models can be categorised into anchored and anchorless object detection algorithms. These algorithms were designed for natural images and were later applied to object detection in remote sensing. They have achieved good results after improvements. Improvements were required because remote sensing images are characterised by scale variability and substantial differences in object shapes, resulting in difficulties in detecting objects in remote sensing images.

We review the literature related to the algorithms used in this paper, including multiscale and high aspect ratioa object detection algorithms and feature enhancement algorithms.

1.1. Multiscale and High Aspect Ratios Object Detection Algorithms

Many object detection algorithms have been proposed to improve the performance of multiscale object detection. For example, the feature pyramid network (FPN)[1] utilises a multiscale pyramid hierarchy inherent in deep CNNs to construct a feature pyramid and creates high-level semantic feature maps at different scales using a top-down architecture with lateral connections. However, the fusion of multilayer feature maps causes errors at the feature scale[2]. Various improvements of this method have been proposed, such as deformable FPN (DFPN)[3], Path Aggregation FPN (PAFPN)[4], and AugFPN[5], and so on. Liu et al.[6]designed an Adaptive Feature Pyramid Network (AFPN) to capture more discriminative features and adaptively combine multiscale features in

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different channels and spatial locations.

Many improvements have been proposed to handle objects with high aspect ratios. For example, anchored algorithms improve the regression results by clustering to obtain similarly shaped anchor boxes. A horizontal bounding box (HBB) may be approximately square, although the object has high acpect ratio. In addition, the complete intersection over union (CIOU) loss[7] and efficient intersection over union (EIOU) loss[8] functions have been used to detect objects with a high aspect ratio. These algorithms typically only consider one of the problems addressed in this study and perform better at small objects or objects with high aspect ratios.

An optimised deep neural network (DNN) was proposed in[9]. It consists of a two-shot neural network with a misalignment localisation strategy to optimise the detection of small and narrow rectangular objects common in high-resolution satellite imagery (HRSI), providing high accuracy for objects with different sizes and aspect ratios. Zhang et al.[10] proposed a multiscale semantic fusion-guided fractal convolution network (MSFC-Net) based on a bidirectional feature pyramid network (BiFPN) that adaptively selects valid details to detect smaller objects. A fractal regression layer adaptively fits a priori information on objects with a wide range of scales and irregular aspect ratios, ensuring the same sampling rate for features in the vertical and horizontal directions.

These algorithms consider multiple scales and high aspect ratios but do not provide sufficient accuracy. Our approach aims to continue exploring the detection of extreme objects.

1.2. Feature Enhancement Algorithm

Commonly used feature enhancement methods utilising deep learning include incorporating an attention mechanism module. Commonly used attention mechanism modules include the self-attention mechanism[11], convolutional block attention mechanism (CBAM)[12], squeeze and excitation network (SENet)[13]. Later improvements were made based on these classic modules. Wang et al. proposed an ECA module based on the SENet channel attention module using a local cross-channel interaction strategy without dimensionality reduction[14]. Hou et al.[15] proposed the coordinate attention mechanism for mobile networks by embedding location information into the channel attention mechanism. This paper uses the CA attention mechanism to significantly enhance performance at a relatively low computational cost.

We propose the one-stage anchorless detector called Attention-Based Multigranular Extreme Object Detection Network (AMEODNet)(Figure.1), which is inspired by CenterNet. In the first part, the feature maps for detecting objects of various sizes are obtained through the multi-granularity information fusion module. In the second part, features for detecting small objects are enhanced using the CA attention mechanism module, while features for detecting medium and large objects are extracted through a parallel asymmetric feature extraction module.

The main contributions and innovations of this paper are summarised as follows:

- A novel one-stage anchorless detector called AMEODNet is proposed to detect small objects and those with high aspect ratios, providing good performance.
- The three parallel asymmetric feature extraction modules are introduced to extract multi-granularity features.
- The CA module is referenced for providing high detection accuracy of objects while using a minimum number of parameters.

The rest of this article is organised as follows. Section 2 presents the details of the proposed methods. Section 3 compares the results of our method with other models on NWPU VHR-10. Section 4 concludes the paper.

2. Proposed Method

This section contains two parts. Part 2.1 introduces the novel network structure shown in Figure.1. Part 2.2 describes the loss function.

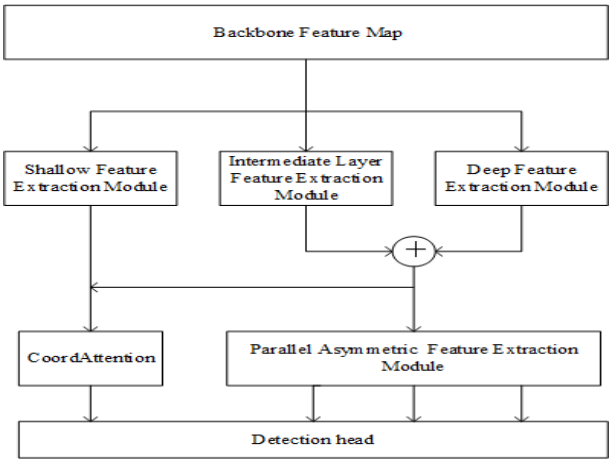


FIGURE 1. Diagram of the AMEODNet structure

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2.1. Network Architecture

The section introduces the network structure, which is divided into three parts. The first part is the fusion module. The second part is the attention mechanism module. The third part is the parallel asymmetric feature extraction module.

2.1.1 Parallel Fractal Fusion Module

The specific structure of this network is shown in Figure.1. The input is a picture X of size 512 × 512 × 3. This image is then passed through the backbone network Resnest[16] for feature extraction to obtain the last five layers of feature maps P_3,P_4,P_5,P_6,and P_7. Finally, these feature maps are fed into a parallel fractal fusion module for feature fusion and detection.

The parallel fusion module primarily includes a shallow feature extraction module, a middle-layer feature extraction module and a deep feature sextraction module. They are two three-layer fractal structures and one two-layer fractal structure, respectively. A fractal fusion module is shown in the Figure.2. Among them, i represents the i-th layer of the fractal structure and m represents mean fusion.The two-layer fractal structure is used for fusing deep features P_6 and P_7. The two three-layer fractal structures are used to fuse features from the previous layer P_3,P_4 and P_5 and intermediate layer features P_4,P_5 and P_6. This setup allows the deep network to extract more accurate detail information[17]. At the same time, the chosen number of layers is aimed at reducing resource consumption.The self-similarity characteristics based on fractals allow the fused features to retain their original properties, which aids in detecting objects of different sizes.

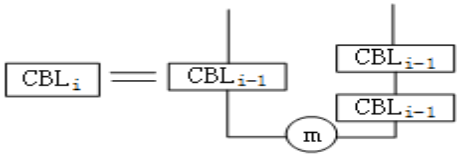


FIGURE 2. Fractal network structure diagram

Among them, the extension of a single convolutional block is shown in Eq. 1. The two-layer fractal structure is an extension of a single convolutional block, as shown in Eq.2. The three-layer fractal structure is an extension of the two-layer fractal structure, as shown in Eq.3.

$$D_{41} = LR(BN(Conv(P_i))) = CBL(P_i). \quad (1)$$

$$D_{42} = CBL(P_{i-1}) + Upsampling(CBL(D_{41})). \quad (2)$$

$$\begin{aligned} D_{43} =& 1/i(CBL(P_{i-1}) + UpsamplingCBL(D_{42}) \\ &+ Upsampling(CBL(CBL(D_{41})))) \end{aligned} \quad (3)$$

In the given formulas, i denotes the i-th layer. $Conv$ represents a convolutional layer. BN indicates a batch normalization layer. RL stands for an activation function. $Upsampling$ represents an upsampling layer.

The parallel fractal fusion module can improve feature extraction at various scales, enhance detail preservation, and increase network capacity to capture complex patterns and relationships in data.

2.1.2 CoordAttention Module

Without significantly increasing complexity, the CoordAttention (CA) mechanism is used to enhance the feature obtained after fusing shallow features, which is cited in[15]. The enhanced features are particularly useful for detecting small objects and those with extreme aspect ratios. The CA module extends the channel attention mechanism from SENet by incorporating positional information encoding, which helps the model more accurately locate and identify the objects of interest.The structure is illustrated in Figure.3.

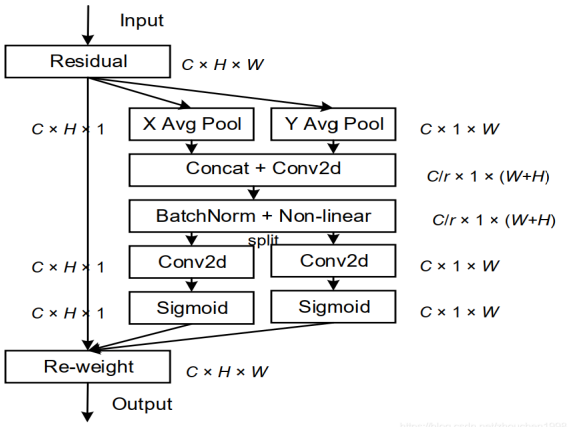


FIGURE 3. CoordAttention mechanism structure diagram

For the feature map X used to detect detail information, pooling kernels of size (H, 1) and (1, W) are applied along the horizontal and vertical directions to encode each channel, generating a pair of directional attention maps. These maps are then concatenated and transformed using a shared 1×1 convolution to produce f. This tensor f is split into two separate tensors f_w and f_h along the spatial dimensions, which are then transformed with two 1×1 convolutions to match the channel count of X, producing g_w and g_h. Finally, these are applied to

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the input feature X. The process is illustrated by the following equation:

$$z_c^h(h) = 1/W * \sum_{1<=i<W} X_c(h,i). \quad (4)$$

$$z_c^w(hw) = 1/H * \sum_{1<=i<H} X_c(j,w). \quad (5)$$

$$f = \delta(F_1[Z^h, z^w]). \quad (6)$$

$$g_h = \sigma(F_h(f_h)). \quad (7)$$

$$g_w = \sigma(F_w(f_w)). \quad (8)$$

$$f_e(i,j) = X * g_c^h(i) * g_c^w(j). \quad (9)$$

2.1.3 Parallel Asymmetric Feature Extraction Module

Since there are some categories with extreme aspect ratio shapes such as bridge and harbor in large objects that are more difficult to detect, the extracted semantic information is feature extracted by using an asymmetric convolution module to fit the shape of the object as shown in Figure.4, which is cited in[18].

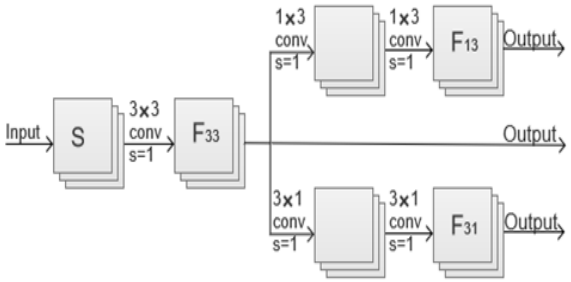


FIGURE 4. Parallel Asymmetric Feature Extraction Module

The feature map obtained by extracting the semantic information branch is first passed through a 3 × 3 convolution operation with a step size of 1 for feature extraction to obtain the feature map . And it is directly fed into the detection head for detection. As shown in Eq.10. Secondly features are extracted by asymmetric convolution for objects that exhibit extreme aspect ratio shapes in the horizontal and vertical directions. In the horizontal direction, two series-connected 1×3 convolution kernels with step size 1 are used to act on to obtain the feature map, where two convolution kernels are used in order to expand the receptive field to 1×5 to facilitate the detection of large objects as shown in Eq.11. For the vertical direction, two convolution kernels of 3 × 1 in series and with a step size of 1 were used to act on to obtain the feature map as shown in Eq.12.

$$F_{33} = Conv(S). \quad (10)$$

$$F_{13} = Conv_{13}(Conv_{13}(F_{33})). \quad (11)$$

$$F_{31} = Conv_{31}(Conv_{31}(F_{31})). \quad (12)$$

Finally, the feature maps f_e, F_{33}, F_{31} and F_{13} are fed to the detection head to update the parameters by computing the loss with the labels and back-propagating.

2.2. Loss Function

Object detection involves recognising the class of the object in the image, determining its positional information and regressing a suitable box to box the object. In order for the detector to perform better, the loss function is defined as follows:

$$Loss = \lambda_{hm} * L_{hm} + \lambda_{off} * L_{off} + \lambda_{wh} * L_{wh} \quad (13)$$

This loss function in Eq.13 is mainly composed of three parts. These three components correspond to the three branches detected by the detector head as heatmap loss, center offset loss and width-height loss, respectively. Whereλ_{hm} ,λ_{off} and λ_{wh} are the weights corresponding to each of the three parts of the loss. In the experiments of this paper, they are set to 1,1 and 0.1.

3. Experiment and Analysis

This section describes the experiment, including the datasets, experimental method, evaluation indicators and analysis results.

3.1. Datasets

The NWPU VHR-10[19] satellite image dataset is used in the experiments of this paper and is often used for object detection in remote sensing images. The NWPU VHR-10 dataset actually consists of a total of 800 images, with 650 images containing objects and 150 images being background images. The object categories include: airplane, ship, storge tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge and vehicle for a total of 10 categories. In this paper, 650 images containing objects are used to randomly divide the training set and test set in the ratio of 5:5 and detect them with horizontal bounding box. The training set is used to train the network to get the weights file. The training set contains 325 images. It is resized to three scales of $*0.4$, $*1$ and $*1.2$ for training. Then it is cropped to $512 * 512$ size sub-images where each sub-image has 200 pixels overlapping. Thus 3637 sub-images can be obtained for training. The test set contains 325 images and each image is scaled to $*0.4$ and $*1$. It is then

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TABLE 1. Experimental results on the NWPU dataset

Method	CenterNet	EGAT	MSFC-Net	EODNet	AMEODNet
Airplane	98.7	100	98.75	98.75	99.18
Ship	75.5	90.86	90.76	92.56	92.55
Storge tank	88.4	99.99	91.98	90.29	91.72
Basebll diamond	99.1	96.84	97.99	98.90	98.54
Tennis court	92.9	90.67	93.35	93.10	96.13
Basketball court	93.5	95.05	98.90	98.56	95.14
Ground track filed	98.3	100	95.78	96.93	96.13
Harbor	93.1	93.67	92.58	94.66	96.37
Bridge	74.4	50.86	64.71	68.41	75.92
Vrhicle	85.7	90.19	93.25	92.30	91.90
mAP	90.0	90.80	91.80	92.44	93.16

cropped to $512 * 512$ size sub-images where each sub-image has 128 pixels overlapping.

3.2. Experimental Details

A ResNeSt-101 network was used for the backbone network. The Adam optimiser was used for the server. Random scaling, cropping, and flipping data enhancement operations were applied to the data. The training batch size was set to 24. The initial learning rate was set to 6.25e-5 and reduced by 0.5 at 50, 90 cycles. All models were implemented by PyTorch 1.11.0 and experiments were executed on RTX 3090 GPUs.

3.3. Evaluation Indicators

In order to quantitatively analyse the performance of the experimental results, some common evaluation metrics are used. For example, accuracy (AP), recall average precision (mAP), etc. The formulas are as follows:

$$recall = TP/(TP + FN). \tag{14}$$

$$precision = TP/(TP + FP). \tag{15}$$

$$mAP = 1/C * \sum_{i=1} AP_i. \tag{16}$$

In the above equation, recall and precision are calculated from TP, FP, TN and FN, which refer to the number of: true positive, false positive, true negative and false negative boxes, respectively. The AP is the integral area under the precision and check-fullness (P-R) curves. mAP denotes the average of the AP sums over the C categories.

Analysis Results In order to verify the effectiveness of AMEODNet, some detectors based convolutional neural network are used on NWPU dataset for comparison experiments. The experimental results are analysed from three perspectives: numerical results such as accuracy, average accuracy and perform visual analysis.

(1)Experiments based on the NWPU dataset are performed on the AMEODNet network and compared with Centernet[20] EGAT[21], MSFC-Net and EODNet. The comparison results are shown in Table 1. Firstly, it can be visualised that the AMEODNet network performs the best in all three categories such as tennis court, harbor and bridge. The accuracies reached 96.13%, 96.37% and 95.92%, respectively. And the average accuracy is the highest, reaching 93.16%. Secondly, we can find that although other categories such as storage tanks and some ships do not have the highest accuracy, they are in the middle accuracy. Moreover, extreme aspect ratio categories such as bridges and harbor have the highest accuracy of 75.92% and 96.37% . Then other networks such as EGAT and MSFC-Net network perform better in multi-scale categories but less well in extreme aspect ratio categories. And CenterNet achieves higher accuracy on large and high aspect ratio objects such as bridges and harbors, but lower accuracy on small objects and small high aspect ratio objects such as storge tanks and ships. The AMEODNet is able to balance these two problems for better detection of extreme objects.

(2)The visualisation results of the experiments based on the NWPU dataset are shown in Figure.5. It can be visualised that the AMEODNet network can accurately detect not only small objects such as vehicle and storge tanks, but it can also accurately detect some objects with extreme aspect ratio shapes such as bridges and ships.

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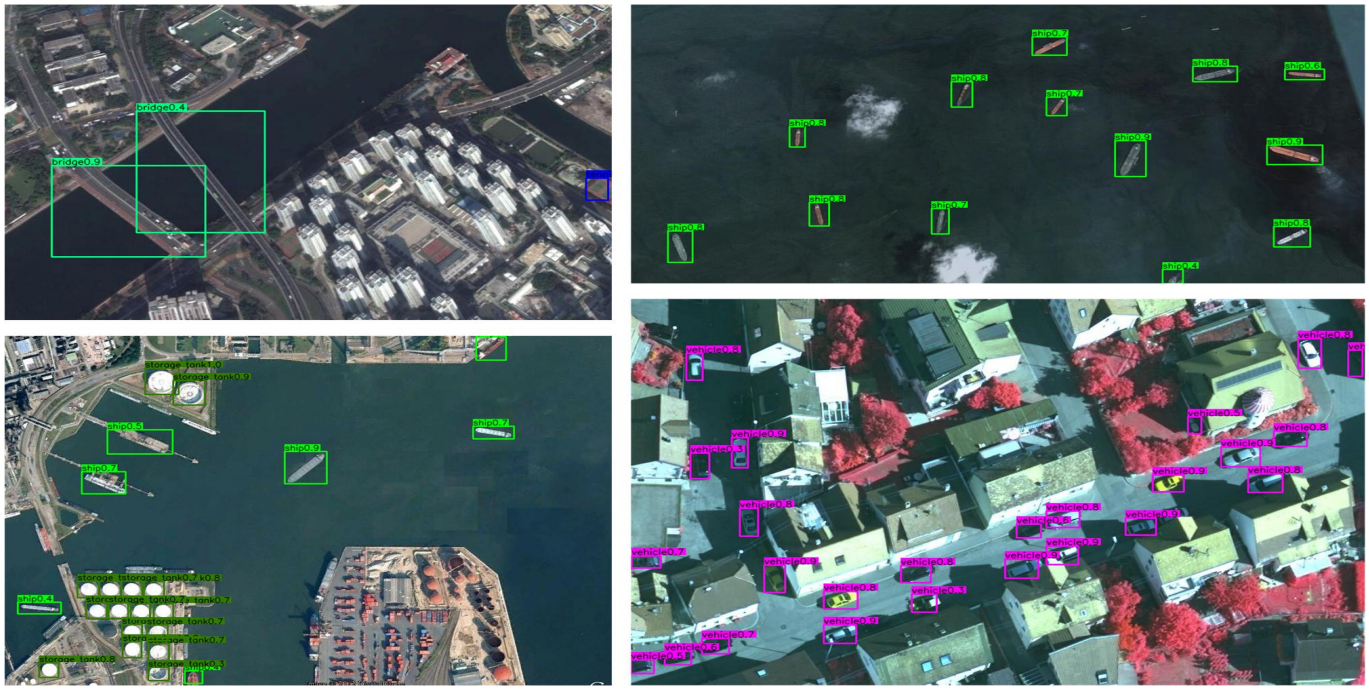


FIGURE 5. Figue of visualisation results on the NWPU dataset

4. Conclusion

In this paper, a single-stage anchorless detector called AMEODNet is proposed in order to solve the problem of multi-granular and very high aspect ratio of objects in remote sensing images. Then, the comparison experiments is performed on the NWPU VHR-10 dataset. The results show that the proposed AMEODNet can obtain better performance than the state-of-the-art detectors. It is demonstrated that AMEODNet can be used to detect extreme objects in the dataset.

However, since AMEODNet follows Centernet using only the center as the positive sample and a very small number of positive samples, it needs to be backed up by using a large amount of data and training rounds to get better results. Then more hardware resources are required for particularly large data. Secondly, the consistency between classification and regression is also an issue that needs further consideration. Addressing these issues will be part of our future work.

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